

THERM-X

Smart Fire and Gas Leakage Detection System Using IoT

Created By:

- Nidhish
- Mohnish
- Vedant
- Rajveer

INDEX

1. Introduction
2. Objectives of the Project
3. Problem Statement
4. Proposed Solution
5. Components Required

Working Principle

1. Circuit Description
2. Software and Programming
3. Advantages
4. Applications
5. Limitations
6. Future Scope
7. Conclusion

1. INTRODUCTION

In today's world, safety has become one of the most important concerns in homes, industries, schools, and offices. Fire accidents and gas leakages can cause severe damage to life and property if not detected at the right time. Many accidents occur because people fail to notice the leakage of harmful gases or the beginning of a fire.

THERM-X is an IoT-based Smart Fire and Gas Leakage Detection System designed to detect dangerous situations quickly and alert people

immediately. The system uses sensors to detect gas leakage and abnormal temperature changes. Once danger is detected, the device activates an alarm and sends information through a Wi-Fi module.

The project combines safety with modern Internet of Things (IoT) technology. It helps in reducing accidents and improving emergency response time. The display module continuously shows the status of the environment, making the system easy to monitor.

This project is affordable, efficient, and suitable for homes, laboratories, industries, restaurants, and public places.

2. OBJECTIVES OF THE PROJECT

The main objectives of THERM-X are:

- To detect fire at an early stage.
- To detect leakage of harmful gases.
- To alert people using an alarm system.
- To display real-time status on a display screen.
- To send alerts using a Wi-Fi module.
- To improve safety using IoT technology.
- To reduce damage caused by fire and gas leakage.
- To create a low-cost and efficient safety system.

3. PROBLEM STATEMENT

Gas leakages and fire accidents are among the major causes of injuries and property damage. Traditional safety systems may not provide immediate alerts or remote monitoring. In many places, people are unaware of dangerous conditions until it becomes too late.

There is a need for a smart system that can:

- Detect danger instantly.
- Provide automatic alerts.
- Work continuously without human supervision.
- Monitor conditions remotely using the internet.

THERM-X solves these problems by using sensors, alarms, displays, and IoT connectivity.

4. PROPOSED SOLUTION

The proposed solution is an IoT-enabled smart safety system that continuously monitors the environment for fire and gas leakage.

The system uses:

- Gas sensors to detect harmful gases.
- Temperature or flame sensors to detect fire.
- A Wi-Fi module for internet connectivity.
- A display screen for status monitoring.
- A buzzer/alarm for emergency alerts.

Whenever the system detects abnormal conditions:

1. The sensor sends data to the microcontroller.
2. The controller analyzes the data.
3. The alarm is activated.
4. Warning messages are shown on the display.
5. Alerts can also be transmitted through Wi-Fi.

This system provides faster response and better protection.

5. COMPONENTS REQUIRED

Hardware Components

Sr. No	Component Name	Purpose
1	Arduino/ESP8266/ESP32	Main controller
2	MQ Gas Sensor	Detects gas leakage
3	Flame Sensor / Temperature Sensor	Detects fire
4	Wi-Fi Module	Sends IoT alerts

Sr. No	Component Name	Purpose
5	LCD/OLED Display	Displays status
6	Buzzer	Emergency alarm
7	LEDs	Visual indication
8	Jumper Wires	Connections
9	Breadboard	Circuit assembly
10	Power Supply	Provides power

Software Requirements

- Arduino IDE
- Embedded C Programming
- IoT Platform (optional)
- Wi-Fi Network

6. WORKING PRINCIPLE

THERM-X works on the principle of sensor-based monitoring and IoT communication.

Step-by-Step Working

1. The gas sensor continuously checks the air for harmful gases.
2. The flame or temperature sensor checks for fire or high heat.
3. Sensor data is sent to the microcontroller.
4. The controller compares the readings with predefined safe limits.
5. If gas leakage or fire is detected:
 - The buzzer starts ringing.
 - LEDs glow as warning indicators.
 - Warning messages appear on the display.
 - The Wi-Fi module sends alerts through the internet.

6. If conditions are normal, the display shows a safe status.

This process runs continuously in real time.

7. CIRCUIT DESCRIPTION

The MQ gas sensor is connected to the analog input pin of the microcontroller. The flame sensor is connected to a digital pin. The buzzer and LEDs are connected as output devices. The display module communicates using I2C or SPI protocol.

The Wi-Fi module allows internet communication and can send notifications to users. The entire circuit operates on a regulated power supply.

The microcontroller acts as the brain of the system and controls all operations.

8. SOFTWARE AND PROGRAMMING

The software is developed using Arduino IDE. Embedded C language is used for programming.

Main Functions of the Program

- Reading sensor values.
- Detecting abnormal conditions.
- Activating alarms.
- Updating the display.
- Sending Wi-Fi alerts.

Algorithm

1. Start the system.
2. Initialize sensors and display.
3. Read gas sensor value.
4. Read flame/temperature sensor value.
5. Compare values with threshold levels.
6. If danger detected:
 - Turn ON buzzer.

- Display warning message.
 - Send Wi-Fi alert.
7. Else:
- Display safe message.
8. Repeat continuously.

9. ADVANTAGES

THERM-X provides many advantages:

- Early fire detection.
- Fast gas leakage detection.
- Real-time monitoring.
- Remote alert system using IoT.
- Low-cost implementation.
- Easy to install.
- Compact and portable design.
- Reduces risk to human life.
- Helps prevent property damage.

10. APPLICATIONS

THERM-X can be used in many places such as:

- Homes and apartments
- Schools and colleges
- Laboratories
- Industries and factories
- Restaurants and hotels
- Gas storage areas
- Shopping malls
- Hospitals

- Offices

11. LIMITATIONS

Despite its advantages, the project has some limitations:

- Requires continuous power supply.
- Internet connection is needed for remote alerts.
- Sensors may require regular calibration.
- Extreme environmental conditions may affect accuracy.
- Limited range depending on Wi-Fi connectivity.

12. FUTURE SCOPE

THERM-X can be improved further in the future by adding advanced features such as:

- Mobile application integration.
- Automatic fire extinguisher system.
- SMS and call alert system.
- AI-based danger prediction.
- Cloud data storage.
- Voice alert system.
- Battery backup for power failure.
- Integration with smart home systems.

These improvements can make the system more intelligent and reliable.

13. CONCLUSION

THERM-X is an efficient IoT-based fire and gas leakage detection system designed to improve safety and reduce accidents. The system continuously monitors environmental conditions and provides immediate alerts during emergencies.

By combining sensors, alarms, display systems, and Wi-Fi communication, the project demonstrates how IoT technology can be used for real-life

safety applications. The project is affordable, practical, and useful in homes, industries, and public areas.

THERM-X successfully achieves its objective of providing early detection and fast warning systems for fire and gas leakage hazards.